

Robust Tangible Projection Mapping with Multi-View Contour-Based Object Tracking

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Figure 1: Robust projection mapping on object grasped and moved by hands.

ABSTRACT

Dynamic projection mapping for moving objects can greatly enhance an augmented reality representation by using projected images because the target object can be freely moved. However, the interaction between the target object and the user has not been sufficiently considered. In this research, we propose tangible projection mapping, with which a user can grasp an object with their hands and move it freely. It uses a simple configuration of two infrared cameras to realize dynamic projection mapping that is robust to hand occlusion.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality

1 INTRODUCTION

Projection mapping is a type of augmented reality technology that can change the appearance of real objects [2]. Images can be projected onto real objects to change their appearance without the need for specialized glasses. To realize accurate image projection using conventional methods, the target object must be stationary. Dynamic projection mapping (DPM) has been developed to overcome this limitation. By using a camera and various sensors to acquire the movement of the target object at high speed and high accuracy, it is possible to perform projection mapping on a moving surface. To maximize the effect of image projection, a method that does not use markers, which are conventionally used for position detection, has been proposed [1, 3, 5].

In the real world, a limited number of objects can move around autonomously. The majority of situations involve a person grasping and moving a real object. For example, when a design mockup onto which various images are projected is being examined in a user's hand, the hand can interfere with the estimation of the position and posture of the object, degrading image projection accuracy. To utilize objects augmented by projection mapping in domains such as education, training, entertainment, and medical care, a method that is robust against hand occlusion is required.

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2 METHOD

In this research, we propose tangible projection mapping (TPM), i.e., DPM for objects that can be freely grasped and moved by users. Fast and accurate object tracking is necessary for DPM. In the present study, we make object tracking robust against hand occlusion.

We adopt a tracking method that uses the object contour acquired by an infrared (IR) camera. Because images are always projected onto the target in DPM or TPM, color-based tracking methods for augmented reality cannot be used. Depth-based methods have been proposed, but they are generally slow. Although high-speed depth-based methods have been proposed, they require special cameras and projectors developed specifically for this purpose [3]. In contrast, the contour-based method adopted here can track objects at 200 Hz or higher using commercially available cameras by matching the object contour obtained from the IR camera with the contour of a previously acquired 3D model of the target object [4].

We now present our method for robustness against hand occlusion. As with other methods, the performance of contour-based methods is greatly degraded by occlusion. Hands can hide the object contour and their own contour can lead to false correspondence points. In situations where color information is unavailable, it is very difficult to recognize hand regions at high speed. Therefore, we introduce the concept of reliability in the correspondence between the object contour obtained from the IR camera and the contour of the 3D model. Object tracking robustness is achieved by eliminating correspondence points that are likely to be false. It is expected that the selection of correspondence points will reduce the total number of points, making tracking more difficult. Therefore, an extra camera is added to obtain contour information from different viewpoints for tracking. The additional information increases the number of correspondence points and decreases pose ambiguity. This use of multiple inexpensive commercially available cameras is an effective and feasible approach.

Figure 2 shows our object tracking process, where the contours obtained from two different viewpoints correspond to the target object and potentially false correspondence points are removed. The final correspondence points are integrated to obtain the differential position and orientation of the target object, $(\Delta x, \Delta y)$, from the previous frame. By repeating this process at high speed, we can achieve high-speed and high-accuracy object tracking.

The correspondence points are selected based on three hypotheses:

- Hypothesis 1: The contour of an occluded region is discontin-

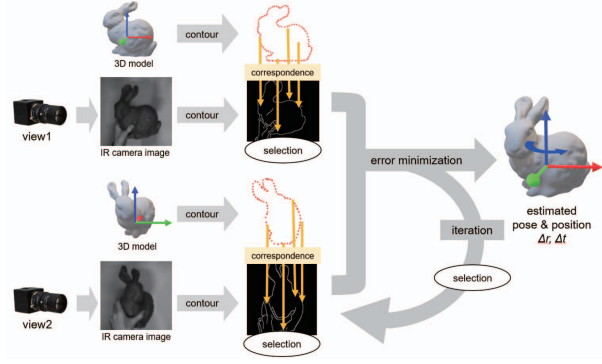


Figure 2: Position and orientation estimation with multiple cameras.

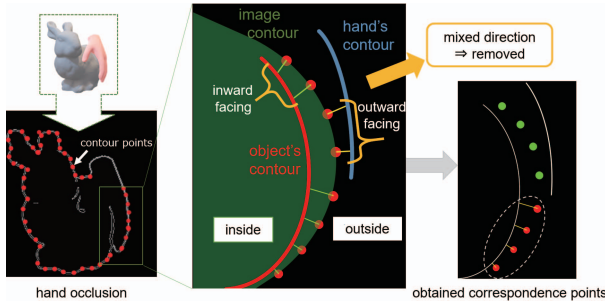


Figure 3: Selection of correspondence points based on Hypothesis 3.

uous.

- Hypothesis 2: In an iterative process that increases estimation accuracy, the distance between appropriate correspondence points decreases with each iteration.
- Hypothesis 3: If the object contour and the hand contour are similar and close to each other, the correspondence directions for the correspondence points will be inconsistent.

Figure 3 shows a specific example of correspondence point selection based on Hypothesis 3.

3 RESULTS

We implemented a prototype system that can realize TPM. Two synchronized IR cameras (resolution: 640×480 pixels; frame rate: 240 frames per second, fps; wavelength: 850 nm) were used for obtaining contour images by using Canny edge detector, and a projector (resolution: 1280×800 pixels; display frequency: 120 Hz) was used for image projection. For tracking and image generation, we used a personal computer (Intel Core i7-8086K, 32 MB of RAM, NVIDIA GeForce RTX 2080Ti) running Windows. The time required for each position and orientation estimation was about 5.0 ms, which was within the projection interval (8.3 ms). The projector was placed above the object, and the two cameras were placed about 1 m away from the object, orthogonally behind and to the left of it.

Figure 1 shows the dynamic image projection results for the proposed method in a situation with a large amount of occlusion. Even when the contour, which is used for object tracking, is half hidden, the image is projected without any significant error.

The translation and rotation errors in Figure 1 are shown in Figure 4. They are compared to the conventional contour-based

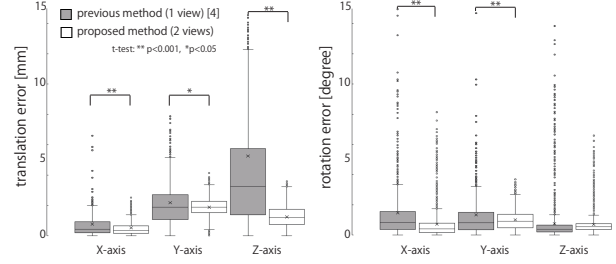


Figure 4: Translation and rotation errors in TPM.

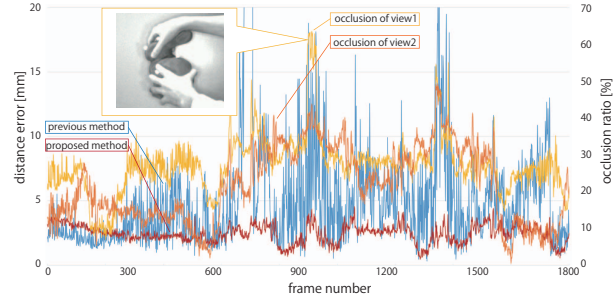


Figure 5: Distance errors and occlusion ratio in TPM.

method using one camera and Hypothesis 1 and Hypothesis 2 [4]. The tracking accuracy was improved in both translation and rotation. The distance error compared to the ground truth obtained from motion capture is shown in Figure 5 along with the occlusion ratio for the contour for the two cameras. Because the conventional method only uses the contour of view1, the error increases with increasing occlusion ratio. In contrast, in the proposed method, the error is kept low even in the situation where the occlusion ratio exceeds 50%.

In the future, we will conduct projections using a wider variety of objects and determine the conditions under which the cameras are effective.

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